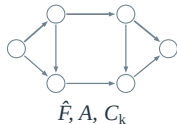


Graph-space price construction

Empirical graph



Joint budget solve

 C_1, b_1 C_2, b_2 C_K, b_K $b_k = B_k/L$ $F_H = 0$

Entropic flow problem

$$\min_F - \langle A, F \rangle + \varepsilon KL(F \parallel \hat{F})$$

$$\text{flow balance} \quad \langle C_k, F \rangle \leq b_k$$

Dual solver

Newton in u Bisection in λ λ^*

Clip + filter

$$\lambda^- \leftarrow (1-\alpha)\lambda^- + \alpha \text{clip}(\lambda^*, 0, c)$$



transitions + costs

Only prices enter the policy update

Policy-space learning

Policy rollout

Episode $\bar{J}$

Cost feedback

$$\beta \leftarrow [\beta + \eta(\bar{J} - B)]_+$$

 β λ^- λ

One priced update

$$\frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO

GAE: A_r, A_c

next rollout

 π_{t+1}